

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

# Mixed

# 03

10 answers · Rationale-rich · Teach from doc

**Q1****D****D. 80**

Choice D is correct. The volume,  $V$ , of a right square prism is given by the formula  $V = s^2h$ , where  $s$  represents the length of the edge of the base and  $h$  represents the height of the prism. It's given that the volume of a right square prism is 2,880 cubic units and the length of the edge of the base is 6 units. Substituting 2,880 for  $V$  and 6 for  $s$  in the formula  $V = s^2h$  yields  $2,880 = 6^2h$ , or  $2,880 = 36h$ . Dividing both sides of this equation by 36 yields  $80 = h$ . Therefore, the height, in units, of the prism is 80.

Choice A is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is 6 units and the volume of the prism is  $144\sqrt{30}$ , not 2,880, units.

Choice B is incorrect. This is the area, in square units, of the base, not the height, in units, of the prism.

Choice C is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is 6 units and the volume of the prism is  $864\sqrt{5}$ , not 2,880, units.

**Q2****D****D.  $100^\circ$** 

Choice D is correct. The sum of the three interior angles in a triangle is  $180^\circ$ . It's given that angle A measures  $50^\circ$ . If angle B measured  $100^\circ$ , the measure of angle C would be  $180^\circ - (50^\circ + 100^\circ) = 30^\circ$ . Thus, the measures of the angles in the triangle would be  $50^\circ$ ,  $100^\circ$ , and  $30^\circ$ . However, an isosceles triangle has two angles of equal measure. Therefore, angle B can't measure  $100^\circ$ .

Choice A is incorrect. If angle B has measure  $50^\circ$ , then angle C would measure  $180^\circ - (50^\circ + 50^\circ) = 80^\circ$ , and  $50^\circ$ ,  $50^\circ$ , and  $80^\circ$  could be the angle measures of an isosceles triangle. Choice B is incorrect. If angle B has measure  $65^\circ$ , then angle C would measure  $180^\circ - (65^\circ + 50^\circ) = 65^\circ$ , and  $50^\circ$ ,  $65^\circ$ , and  $65^\circ$  could be the angle measures of an isosceles triangle. Choice C is incorrect. If angle B has measure  $80^\circ$ , then angle C would measure  $180^\circ - (80^\circ + 50^\circ) = 50^\circ$ , and  $50^\circ$ ,  $80^\circ$ , and  $50^\circ$  could be the angle measures of an isosceles triangle.

**Q3****9**

The correct answer is 9. It's given that the equation  $y = px + r$  defines the line. In this equation,  $p$  represents the slope of the line and  $r$  represents the  $y$ -coordinate of the  $y$ -intercept of the line. It's given that the line has a slope of 9. Therefore, the value of  $p$  is 9.

## Q4

D

D. 44

Choice D is correct. It's given that  $f(x) = x + 6$ . Substituting 2 for  $x$  in the function  $f$  yields  $f(2) = 2 + 6$ , or  $f(2) = 8$ . It's also given that  $g(x) = 6x$ . Substituting 2 for  $x$  in the function  $g$  yields  $g(2) = (6)(2)$ , or  $g(2) = 12$ . Substituting 8 for  $f(2)$  and 12 for  $g(2)$  in the expression  $7f(2) - g(2)$  yields  $7(8) - 12$ , or 44. Therefore, the value of  $7f(2) - g(2)$  is 44.

Choice A is incorrect. This is the value of  $f(2) - g(2)$ , not  $7f(2) - g(2)$ .

Choice B is incorrect. This is the value of  $f(2)$ , not  $7f(2) - g(2)$ .

Choice C is incorrect and may result from conceptual or calculation errors.

## Q5

B

B.  $k = 4.2x$

Choice B is correct. The relationship between  $x$  food calories and  $k$  kilojoules can be modeled as a proportional relationship. Let  $(x_1, k_1)$  and  $(x_2, k_2)$  represent the values in the first two rows in the table:  $(4.0, 16.7)$  and  $(9.0, 37.7)$ . The rate of change, or  $\frac{(k_2 - k_1)}{(x_2 - x_1)}$ , is  $\frac{21}{5} = 4.2$ ;

therefore, the equation that best represents the relationship between  $x$  and  $k$  is  $k = 4.2x$ .

Choice A is incorrect and may be the result of calculating the rate of change using  $\frac{(x_2 - x_1)}{(k_2 - k_1)}$ .

Choice C is incorrect because the number of kilojoules is greater than the number of food calories. Choice D is incorrect and may be the result of an error when setting up the equation.

**Q6****3600**

The correct answer is 3,600. The predicted number of bacteria initially measured in the growth medium is the value of  $y$  when  $x = 0$ . Substituting 0 for  $x$  in the given equation yields  $y = 3,600(a)^0$ . Since  $a$  is positive, this equation is equivalent to  $y = 3,600(1)$ . Therefore, when  $x = 0$ , the value of  $y$  is 3,600. Thus, the predicted number of bacteria initially measured in the growth medium is 3,600.

**Q7****C**

C.  $px = 2,0001.03^x$

Choice C is correct. It's given that a function  $p$  estimates that there were 2,000 animals in a population in 1998 and that each year from 1998 to 2010, the number of animals in this population increased by 3% of the number of animals in the population the previous year. It follows that this situation can be represented by the function  $px = a1 + \frac{r}{100}^x$ , where  $px$  is the estimated number of animals in the population  $x$  years after 1998,  $a$  is the estimated number of animals in the population in 1998, and each year the estimated number of animals increased by  $r\%$ . Substituting 2,000 for  $a$  and 3 for  $r$  in this function yields  $px = 2,0001 + \frac{3}{100}^x$ , or  $px = 2,0001.03^x$ .

Choice A is incorrect. This function represents a population in which each year the number of animals increased by 200%, not 3%, of the number of animals in the population the previous year.

Choice B is incorrect. This function represents a population in which each year the number of animals increased by 97%, not 3%, of the number of animals in the population the previous year.

Choice D is incorrect. This function represents a population in which each year the number of animals decreased, rather than increased, by 3% of the number of animals in the population the previous year.

**Q8****A**

A.  $\frac{h^{10}}{q^{14}}$

Choice A is correct. For positive values of  $a$ ,  $\frac{a^m}{a^n} = a^{m-n}$ , where  $m$  and  $n$  are integers. Since it's given that  $h > 0$  and  $q > 0$ , this property can be applied to rewrite the given expression as  $h^{15-5}q^{7-21}$ , which is equivalent to  $h^{10}q^{-14}$ . For positive values of  $a$ ,  $a^{-n} = \frac{1}{a^n}$ . This property can be applied to rewrite the expression  $h^{10}q^{-14}$  as  $h^{10}\frac{1}{q^{14}}$ , which is equivalent to  $\frac{h^{10}}{q^{14}}$ .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q9****C**

C. 2, 4, 8, 16, 32, 64, 128, 256, 512

Choice C is correct. If the values in a data set are ordered from least to greatest, the median of the data set will be the middle value. Since each data set in the choices is ordered and contains exactly 9 data values, the 5th value in each is the median. It follows that the median of the data set in choice C is 32. The sum of the positive differences between 32 and each of the values that are less than 32 is significantly smaller than the sum of the positive differences between 32 and each of the values that are greater than 32. If 32 were the mean, these sums would have been equal to each other. Therefore, the mean of this data set must be greater than 32. This can also be confirmed by calculating the mean as the sum of the values divided by the number of values in the data set:  $\frac{2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + 512}{9} = 113\frac{5}{9}$ .

Choices A and B are incorrect. Each of the data sets in these choices is symmetric with respect to its median, so the mean and the median for each of these choices are equivalent. Choice D is incorrect. The median of this data set is 207. Since the sum of the positive differences between 207 and each of the values less than 207 is greater than the sum of the positive differences between 207 and each value greater than 207 in this data set, the mean must be less than the median.

## Q10

D

D. Approximately 30% of all largemouth bass in the pond weigh more than 2 pounds.

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Choice D is correct. The sample of 150 largemouth bass was selected at random from all the largemouth bass in the pond, and since 30% of the fish in the sample weighed more than 2 pounds, it can be concluded that approximately 30% of all largemouth bass in the pond weigh more than 2 pounds.

Choices A, B, and C are incorrect. Since the sample contained 150 largemouth bass, of which 30% weighed more than 2 pounds, this result can be generalized only to largemouth bass in the pond, not to all fish in the pond.

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### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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